

Supervised Bayesian Matrix Factorization Identifies Prognostic Gene Expression Programs

Presentation Notes

Andrew Walther

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This companion is a preparation guide for an approximately 15–20 minute lab meeting talk. This deck is built directly from the 6/18 template but is **not a diff against it** — it presents the current state of the method, not a changelog. Two changes run through the entire talk: the loadings predictor ($L\beta$) is dropped from the narrative entirely (only the projection predictor $(YF)\beta$ is presented), and K selection is a **two-stage split** — a consensus of ELBO, BIC, log-likelihood, and cross-validated external C-index picks K_{init} ; ARD shrinkage then determines K_{eff} from that one fit, with no separate per- K validation loop for K_{eff} itself. Cross-validation is *not* gone; it is one of four inputs to K_{init} , not the sole method, and it no longer determines K_{eff} on its own.

Executive Summary

Pancreatic ductal adenocarcinoma (PDAC) needs validated molecular prognosis, but the conventional *two-step* pipeline — unsupervised factorization, then test factors for survival after the fact — chases the largest expression variance and misses low-variance prognostic programs. This work is a *joint, supervised Bayesian matrix factorization*: the projection predictor $\eta = (YF)\beta$ scores new patients with the identical formula used at training, so external scoring is exact and leakage-free.

- **K selection, reframed:** K_{init} is chosen from a consensus of ELBO, BIC, an ELBO-style joint log-likelihood, and cross-validated external C-index, swept over $K_{\text{init}} = 2..20$. The four criteria *disagree* — ELBO/BIC prefer $K_{\text{init}} = 3$, external C-index prefers $K_{\text{init}} = 12$ — and that disagreement is presented honestly, with $K_{\text{init}} = 7$ kept as the practical recommendation (stable $K_{\text{eff}} = 2$, externally competitive, reachable from a single fit).
- **Findings:** On real PDAC data (train TCGA-PAAD + CPTAC, validate five held-out cohorts) the model attains mean external C = [0.627](#), beating the fairest independent two-step baseline (EBMF $\rightarrow$ Cox, $K = 40$, LASSO stage 2) by a pooled bootstrap [+0.026](#) (95% CI [0.0002, 0.0498], significant). A signal-strength sweep shows the joint model matches the unsupervised baseline when there is no survival signal to exploit, and pulls ahead as signal grows.
- **Simulation, re-run under ARD:** the multi-cohort simulation now selects K the way real data does (over-specified K_{init} + ARD pruning) instead of an oracle true- K . Recovery, specificity, and C-index are flat across K_{init} (15 seeds) — over-specifying K costs nothing. A secondary false-positive-rate question trends favorably but is **not statistically significant** ($p = 0.07$) — reported as a trend, not a finding.

- **Parsimony:** of the programs $K_{\text{init}} = 7$ keeps, only 2 carry survival signal — an adverse basal-like program and a protective classical program, both now biologically annotated by 5 independent methods; the 2 genomics-only kept programs are newly characterized too (stroma, immune infiltrate).

Opening

Slide numbers below match the deck’s on-screen numbering (title slide = 1). The three section-divider slides (3 Background, 5 Methods, 12 Results) count toward that numbering; appendix slides are “un-counted” in the deck and are labeled A1–A3 here.

Pacing. 15–20 minutes over 24 slides is under one minute per slide on average — the appendix (A1–A3) is Q&A-only, so budget the pace against the 24 counted slides, not 27. Text in **navy bold** is a must-say point; underlined Carolina blue marks headline numbers.

Slide 1 — Title Slide

Script. Thanks everyone — a project update on the supervised Bayesian matrix factorization work. Since June, two things changed: **K selection no longer relies on cross-validation alone** — it is now a consensus of fit criteria plus ARD shrinkage — and **only the projection predictor is presented going forward**. Frame the talk as a methods update and a validated results update.

- **Main Takeaway:** This deck describes the current state of the method, not a list of changes since June — avoid “here’s what’s new” framing throughout.
- **Timing:** Two sentences, then the agenda.

Main Deck

Slide 2 — Agenda

Script. A couple of minutes on motivation, then the method — most of it is the two-stage K -selection framework — then **most of the talk is results:** the K_{init} sweep, gene programs and survival, the joint-vs-two-step comparison, and the multi-cohort simulation. Close with impact and next steps.

- **Main Takeaway:** Results-forward talk; the K -selection story is the methodological thread running through everything.

Slide 3 — Section divider: Background

Script. (Divider — no content.) Pause to transition from agenda into motivation.

Slide 4 — Background: PDAC needs molecular prognosis

Script. PDAC has five-year survival under 12% and no adopted molecular stratification at diagnosis. Goal: recover **gene expression programs** — a column of F with patient activation in L — that

predict survival, validated on independent cohorts. Why jointly rather than two-step: unsupervised axes chase the largest variance, often batch; **a 15%-variance program may be batch while a 3% one separates survivors**. Supervision steers factors toward survival-relevant structure during estimation.

- **Main Takeaway:** Genuine clinical need; joint-over-two-step is the thread of the whole talk.
- **Good Line to Stress:** “A high-variance program can be batch effect; a low-variance program can be the one that separates survivors. Supervision prioritizes the second.”

Slide 5 — Section divider: Methods

Script. (Divider — no content.) Signal the shift into method.

Slide 6 — The model — two coupled components

Script. Two coupled pieces: $Y = LF^\top + E$ for expression, $h_i(t) = h_0(t) \exp(\eta_i)$ for survival, coupled through the **shared gene-program weights** F . The table defines every symbol, including $\eta_i = (Y_i F) \beta$ — the projection predictor is the only one shown now. A program is **prognostic only** if $\beta_k \neq 0$. Baseline hazard $h_0(t)$ is non-parametric — cancels in the partial likelihood, no shape assumption.

- **Main Takeaway:** Shared F drives both genomics reconstruction and the survival score; only the projection predictor is discussed from here on.
- **Good Line to Stress:** “ $\beta_k = 0$ means the model keeps that program for expression but not for prognosis — that separation is the whole point.”

Slide 7 — The model: projection predictor and explicit priors

Script. The projection predictor $\eta = (YF)\beta$ scores a new patient with the **identical training formula** — exact, leakage-free, no OLS re-projection approximation. Priors, stated explicitly: **point-exponential (non-negative) on L, F , normal on β** , closed-form MLE for τ — **not point-normal**. The normal prior on β is what makes **ARD-based K selection possible** two slides from now — shrinkage toward zero, not a hard spike-and-slab switch.

- **Main Takeaway:** Exact external scoring is why this is the recommended (and now only) linear predictor.
- **Emphasis:** Flag the normal-on- β detail explicitly — it is the mechanism the K -selection slide relies on.
- **Likely Question:** “Why drop $L\beta$ entirely — wasn’t it the fully-joint reference?” It still exists in the codebase, but the projection model has always been the recommendation (exact scoring, no re-projection mismatch), and simplifying the narrative to one predictor keeps focus on the results that matter for this update.

Slide 8 — The objective — ELBO & variational family

Script. Mean-field variational family; CAVI maximizes the **ELBO** — Gaussian genomics fit, a 2nd-order Taylor Cox survival term (keeps it conjugate), and a per-block KL term (the adaptive shrinkage). Shared F in both likelihood terms *is* the supervision. **No α to tune** ($\lambda = 1$). New since June: the ELBO is now **also one of the four K_{init} -selection criteria**, not excluded from it.

- **Main Takeaway:** Supervision is one joint objective; the ELBO now does double duty as fitting objective *and* one K_{init} -selection input.
- **Likely Question:** “Doesn’t using the ELBO to fit *and* to help pick K double-count?” No — fitting happens at a fixed K_{init} ; comparing ELBO *across* different K_{init} fits afterward is a separate, standard model-comparison use of the same quantity, analogous to comparing log-likelihoods across models with different parameter counts (which is exactly why BIC, which penalizes for that, is shown alongside it).

Slide 9 — Inference — factor-wise CAVI

Script. Coordinate-ascent, one factor at a time, each update an EBNM sub-problem. β_k ’s precision is governed by its own projection score’s variance, so a **no-signal factor shrinks** $\beta_k \rightarrow 0$ **on its own** — the mechanism ARD relies on. Convergence: relative ELBO change below 10^{-5} ($\Delta L, \Delta \beta$ are diagnostics only). Sign correction: orient by **training** concordance, reuse at test.

- **Main Takeaway:** One factor at a time; convergence is judged on the ELBO itself.
- **Good Line to Stress:** “A factor with no survival signal shrinks its own β toward zero as a natural consequence of the update, not because of an external rule — that’s ARD.”

Slide 10 — Preprocessing is crucial — per-platform normalization

Script. Unchanged from June, still the most important practical finding. $A_k = \sum_i w_i (YF)_{ik}^2$; without per-platform z-standardization, between-platform variance swamps A_k and $\beta \rightarrow 0$. **10 of 12** non-per-platform pipelines collapse in an 18-config benchmark. No leakage — normalization fit within each training cohort before any split.

- **Main Takeaway:** Per-platform normalization before merging is the difference between a working model and $\beta \rightarrow 0$.
- **Likely Question:** “Isn’t per-cohort normalization leakage?” No — fit within each training cohort before any split; held-out cohorts get the same per-cohort recipe at scoring time, never using training or test labels.

Slide 11 — Selecting the number of programs: a two-stage framework

Script. The core methods update. **Stage 1** picks K_{init} from a consensus of four criteria: ELBO, BIC ($\text{df} = K_{\text{init}} \cdot (n + p + 1)$, charged by starting K , not K_{eff}), an ELBO-style log-likelihood, and cross-validated external C-index — **these need not agree**, and the full curve is shown in Results, not just the answer. **Stage 2** is ARD: at the chosen K_{init} , ‘classify_factors()’ splits factors into

survival-active ($|\hat{\beta}| > 0.001$), genomics-only ($\text{PVE} > 1\%$), and dead. Running example: $K_{\text{init}} = 7 \rightarrow$ [2 survival-active](#) + [2 genomics-only](#) (4 kept, 3 pruned).

- **Main Takeaway:** Two stages, not one CV loop — CV-based C-index informs K_{init} alongside three other criteria; ARD (not CV) determines K_{eff} .
- **Emphasis:** Say explicitly that cross-validation was *not* removed — it is one of four K_{init} inputs. What changed is that it no longer single-handedly determines K_{eff} .
- **Likely Question:** “Are the ARD thresholds ($\text{PVE} > 1\%$, $|\hat{\beta}| > 0.001$) principled?” Be honest: no. `beta_threshold` was calibrated to this model’s own observed $|\hat{\beta}|$ scale (roughly 0.003–0.008 under YFB), not drawn from a citable sparse-factor-model source; `pve_threshold` is an uncited round number. A literature check and a sensitivity sweep are open items (ROADMAP.md) — flagged as a genuine limitation, not glossed over. See also the Limitations slide.

Results

Slide 12 — Section divider: Results

Script. (Divider — no content.) Signal the shift into results; most of the remaining time lives here.

Slide 13 — Data: Synthetic Validation & PDAC Cohorts

Script. Two datasets. Simulation: EBMF-grounded ground truth, shared vs. study-specific factors, three scenarios, now **two arms** (the projection model, $\text{EBMF} \rightarrow \text{Cox}$) fit at an over-specified K_{init} pruned by ARD, mirroring the real-data procedure. Real data: train TCGA-PAAD + CPTAC ($n \approx 273$), validate on five fully held-out cohorts, scored by the exact projection formula.

- **Main Takeaway:** The simulation now tests the *actual* K-selection procedure, not an oracle short-cut.

Slide 14 — PDAC Cohorts: Prognostic Gene Programs

Script. Gene-weight heatmap, but note the source changed: this now shows **only the 4 ARD-kept factors** (from the 8/21 progress-book figure), not all 7 factors from the June deck’s heatmap. $K_{\text{init}} = 7 \rightarrow$ 2 survival-active (Programs 3, 7) + 2 genomics-only (Programs 5, 6: **desmoplastic stroma and immune infiltrate** — newly characterized this cycle) kept. The largest expression program (collagen/ECM/stromal, Program 5) is *not* survival-active.

- **Main Takeaway:** Supervision separates expression-only structure from prognostic structure; all 4 kept programs are now biologically named, not just the 2 survival-active ones.
- **Likely Question:** “Why does the heatmap only show 4 factors, not all 7?” The 3 fully-pruned (dead) factors carry no real gene-program structure to show — ARD’s own classification says so, and the 8/21 pathway-enrichment work confirmed it (zero enriched gene sets for the dead factors, independently corroborating the split).

Slide 15 — PDAC Cohorts: Prognostic Association

Script. Unchanged from June. Split at median projection score $(YF)_k$: Program 7 adverse (MET, ITGA3, glycolytic genes), Program 3 protective (epithelial differentiation markers). Schoenfeld test: PH holds.

- **Likely Question:** “Does the joint β sign always match the direction you report?” Not necessarily — Program 7’s joint coefficient can run opposite its marginal adverse direction (suppression among correlated programs). We label by the empirical KM direction and stratify by (YF) projection, not raw β sign. Deliberate, not an inconsistency.

Slide 16 — Choosing K_{init} : the consensus sweep

Script. The full $K_{\text{init}} = 2..20$ sweep, new this cycle. **ELBO and BIC both prefer $K_{\text{init}} = 3$** ; BIC’s complexity penalty ($\approx 13,100$ per unit K_{init} here) dominates past single digits. **External C-index prefers $K_{\text{init}} = 12$** , though $K_{\text{init}} = 7$ through 20 mostly plateau around $C \approx 0.627$ (two isolated dips at $K_{\text{init}} = 11, 13$, consistent with a known single-seed CAVI factor-collapse artifact, not new). $K_{\text{init}} = 7$ kept as the recommendation: stable $K_{\text{eff}} = 2$, simplest single-fit path.

- **Main Takeaway:** The four criteria genuinely disagree — say so plainly, don’t paper over it.
- **Good Line to Stress:** “I’m showing you the actual curves, not just the recommendation, because the disagreement itself is worth discussing.”
- **Likely Question:** “Why not just use $K_{\text{init}} = 3$ if ELBO/BIC prefer it?” $K_{\text{init}} = 3$ ’s external C-index (0.594) is well below the $K_{\text{init}} \geq 7$ plateau (0.627) — BIC’s penalty doesn’t know about held-out generalization, only in-sample fit complexity. We weight external validity heavily, which is why $K_{\text{init}} = 7$, not 3, is the practical recommendation.
- **Likely Question:** “Why not $K_{\text{init}} = 12$, since external C prefers it?” It’s a legitimate alternative — $K_{\text{init}} = 7$ through 20 are close on the plateau. $K_{\text{init}} = 7$ is kept for simplicity (single fresh fit) and because K_{eff} is identical (2) across that whole range, so nothing about the biological conclusion changes with the choice.

Slide 17 — Joint model vs. the unsupervised two-step

Script. Real-data analog of the synthetic comparison. Two arms: the projection model vs. the **fairest** independent two-step baseline (EBMF $\rightarrow$ Cox, $K = 40$, LASSO stage 2 — large, YFB-uninformed K , regularized stage 2, not a small K or an overfitting plain `coxph()`). Projection mean C = **0.627** vs. EBMF **0.601**; pooled bootstrap difference **+0.026, 95% CI [0.0002, 0.0498], significant** ($n = 616$ pooled).

- **Main Takeaway:** This is the hardest-to-argue-with two-step comparison available, not a favorable cherry-pick.
- **Emphasis:** This CI was independently re-derived from paired risk scores before this talk, not quoted from memory — say so if pressed on where the number comes from.
- **Likely Question:** “Why $K = 40$ for the baseline, not something smaller?” Larger K gives the unsupervised step more capacity to find whatever structure exists, without ever seeing survival

outcomes — handing it a *smaller* K (like 7, matched to YFB) actually hands over part of the answer, since the joint model chose that K using its own selection procedure. $K = 40$ with a regularized (LASSO) stage-2 Cox is the configuration built specifically to be the fairest test, not the one most favorable to us.

Slide 18 — Joint model vs. two-step as survival signal strength varies

Script. Reused from the 8/21 progress-book work, unchanged this cycle. One figure, two points: at zero simulated survival signal, **joint** $\approx$ **EBMF-only** — no fabricated prognostic signal. As signal strength increases, the joint model pulls ahead.

- **Main Takeaway:** Supervision is a free option — costs nothing when uninformative, helps when there is real signal.

Slide 19 — Synthetic Results: ARD matches oracle- K performance

Script. New this cycle: the simulation now selects K the way real data does. Recovery, specificity, and C-index are **flat across** $K_{\text{init}} = 6 \rightarrow 12 \rightarrow 20$ in every scenario, with 15 seeds (up from 5, for a properly powered comparison). $K_{\text{init}} = 6$ (the oracle true K) is kept only as an internal reference point, not a claim that we know the true K on real data.

- **Main Takeaway:** Over-specifying K and trusting ARD costs nothing relative to knowing the true K in advance — the core validation this simulation exists to provide.
- **Likely Question:** “Why only 2 arms now, not 5 like June?” $L\beta$ and the cohort-indicator arms are dropped from the whole narrative (see Slide 7); June already found no meaningful YFB-vs-YFB+cohort difference, so re-testing it here would add nothing new.

Slide 20 — Synthetic Results: false-positive rate, a trend not yet significant

Script. A secondary question: does over-specifying K_{init} increase false-positive survival-active calls on non-signal factors? At 5 seeds it looked like a clean improvement; **at 15 seeds it softened** — hybrid scenario $0.35 \rightarrow 0.22 \rightarrow 0.27$, paired t -test $K = 6$ vs. 12 gives $p = 0.07$: directionally consistent, **not conventionally significant**. Negative-control scenario is flat at 0.10 across all K_{init} — the apparent 5-seed improvement there was noise.

- **Main Takeaway:** Report this as what it is — a promising trend, not a confirmed finding. Increasing to 15 seeds is itself worth mentioning as good practice, not something to skip past.
- **Good Line to Stress:** “A small, non-zero coefficient on a non-signal factor doesn’t meaningfully move the risk score in practice, which is consistent with what we see here even when a false positive occurs.”
- **Likely Question:** “Why show a non-significant result at all?” Because the direction is consistent and worth tracking, and because showing it with an honest p -value is more credible than either hiding it or overstating it as confirmed.

Slide 21 — Model Comparison Summary

Script. Two rows now, not four: the projection model vs. the unsupervised two-step. Table note explains K_{init} (consensus-selected) vs. K_{eff} (ARD-determined) explicitly.

- **Main Takeaway:** Recommendation unchanged in substance (projection predictor, per-platform z-std, no cohort indicator) — only the comparison table’s row count and the K -selection description changed.

Slide 22 — Impact / key findings

Script. Five points: joint beats two-step at no cost when uninformative; preprocessing is decisive; K selection is the two-stage split (say it exactly the same way as Slide 11 — CV informs K_{init} , ARD determines K_{eff}); over-specifying K_{init} costs nothing (simulation, 15 seeds); mean external C = [0.627](#). Then limitations, stated plainly: the four K_{init} criteria disagree; the ARD thresholds aren’t literature-derived; small cohorts are noisy; biological annotation is done, DeSurv head-to-head is next.

- **Main Takeaway:** Every limitation named here has already been said once earlier in the talk — this slide is a recap, not new information, which is exactly the point: nothing is being saved for later or hidden.

Slide 23 — Future directions & summary

Script. Current status, not “addressed since June” framing. Next steps: lead with **multiple cohorts / multiple modalities via indicator columns in L** — stack methylation + expression, possibly modality-specific priors, leave-one-study-out and train-on-A-vs-B-vs-A+B validation. Then manuscript prep. Summary panel restates the recommended configuration with numbers.

- **Main Takeaway:** The cohort-indicator idea from June is repositioned here as a forward-looking direction (multi-cohort/multi-modality), not a tested-and-rejected result — don’t imply it was tried and failed.

Slide 24 — Discussion

Script. Open the floor. Appendix covers the full K_{init} sweep table, convergence diagnostics, and the DeSurv comparison.

Anticipate These Audience Questions

- **“Didn’t you say cross-validation was replaced by ARD? Isn’t that inconsistent with C-index still being on the K-sweep slide?”** No inconsistency: cross-validated external C-index is one of four inputs that help choose K_{init} (Stage 1). What ARD replaced is the use of cross-validation to determine K_{eff} (Stage 2) directly — that used to require a second validation loop; now it falls out of a single over-specified fit.
- **“The four K_{init} criteria disagree by a lot – doesn’t that undermine the method?”** It’s an honest finding, not a flaw to hide: BIC penalizes model complexity in a way that doesn’t

know about held-out generalization, so it favors small K_{init} ; external C-index does know about generalization and prefers more capacity. The biological conclusion ($K_{\text{eff}} = 2$ survival-active) is stable across the whole $K_{\text{init}} = 7\text{--}20$ plateau regardless of which criterion wins, which is the more important robustness check.

- **“Are the ARD retention thresholds arbitrary? Would different thresholds change your programs?”** They are not literature-derived — `beta_threshold=0.001` was calibrated to this model’s own coefficient scale, and `pve_threshold=0.01` is an uncited round number. Shifting them would plausibly move which factors classify as survival-active vs. genomics-only, which is exactly why a literature check and sensitivity sweep are tracked as open items rather than treated as settled.
- **“Why did the two-step comparison number change from what I remember hearing before (0.581 or similar)?”** That was an earlier, less carefully regularized baseline (smaller K , plain `coxph()` that overfits at large K). The current [0.601](#) baseline uses $K = 40$ with a LASSO-regularized stage-2 Cox specifically to remove that overfitting artifact and give the unsupervised approach every reasonable advantage before comparing.
- **“Why present a non-significant result (the false-positive-rate trend) at all?”** Because the direction is consistent across 15 seeds even though the test isn’t significant, and stating the p -value plainly is more credible than either omitting the result or overstating it as confirmed. It’s also directly relevant to a concern raised in an earlier meeting about ARD over-counting survival-active factors — this is the follow-up data on that concern.
- **“How does this compare to the lab’s penalized supervised-MF (DeSurv) method?”** Bayesian counterpart: same supervised NMF + Cox idea, same TCGA + CPTAC training and five external cohorts, same projection-based scoring, borrowed combined-rank gene selection. Differences: DeSurv tunes k, α, λ jointly by Bayesian optimization; this model selects K_{init} by the four-criteria consensus and lets ARD prune, with empirical-Bayes priors learning shrinkage (no α, λ to tune) and posterior uncertainty. Different primary metrics reported (pooled HR/SD vs. mean external C), so a matched same-protocol head-to-head remains a planned next step.
- **“What happened to the cohort-membership indicator from the June deck?”** Not dropped as a failed idea — repositioned as a forward-looking multi-cohort/multi-modality direction (Future Directions), since the more useful framing now is extending L to combine multiple cohorts and/or modalities (e.g. methylation + expression), not re-litigating June’s single-modality result.

Appendix Slide Scripts

Slide A1 — Appendix: K_{init} sweep, full table

Script. If someone wants the actual numbers behind the sweep curves, this is the table — all 19 K_{init} values with survival-active/genomics-only/dead counts, ELBO, BIC, log-likelihood, and external C, $K_{\text{init}} = 7$ highlighted.

- **Main Takeaway:** Full transparency on the sweep — nothing on the curve slide is cherry-picked.

Slide A2 — Appendix: Convergence & survival-activity magnitudes

Script. Unchanged from June. RMSE stabilizes to a plateau while the ELBO — the actual objective — governs stopping (relative change below 10^{-5}). The $|\hat{\beta}_k|$ bars show the shrinkage: most programs fall below threshold, a couple clear it.

- **Main Takeaway:** Convergence is judged on the ELBO; K_{eff} is read off the few programs with $|\hat{\beta}|$ above threshold.

Slide A3 — Appendix: Relation to DeSurv

Script. Backup for the inevitable comparison question (see also the Anticipated Questions section above). Be collegial and precise: Bayesian sibling of DeSurv, same data and scoring, genuine differences are adaptive priors and fewer tuned knobs, plus posterior uncertainty. Different primary metrics reported; matched head-to-head is the honest next step, not a superiority claim today.

- **Main Takeaway:** Same problem and data, Bayesian estimation; head-to-head is planned, not done.